

Model Purpose

The PRM fleet optimisation model is a flight-level mixed integer optimisation model used to assess the operational requirement for PRM service vehicles and staff across a schedule period.

The model assigns each flight to one mutually exclusive service mode, assigns physical Ambulifts and minibuses where required, checks that assigned vehicles do not overlap in time, allows future minibus candidates to be purchased where required, and produces reporting outputs on fleet utilisation, fleet requirement, service assignments, shortfalls in vehicles, SLA performance and staffing.

The model is designed around the following principles:

- The model operates at flight level
- Service timings are represented through precomputed fixed intervals
- Existing Ambulifts are fixed and can not be purchased
- Existing minibuses are fixed
- Future minibuses may be purchased
- Passenger demand and scenario shaping, including 90th and 100th percentile demand assumptions are prepared upstream before the optimiser is called.

The model is to answer:

Given the flight demand, service-mode rules, physical vehicle fleet, future minibus candidates, vehicle capacities, fixed service intervals and operational constraints, which service mode and vehicle assignments minimise shortfalls in vehicle numbers, future purchases, current vehicle usage, staff-minutes, excess assignments and arrival SLA risk?

Model Scope

1. Set and Indices

- F : set of all flights in the schedule, indexed by f . This is subdivided into F_{arr} (arrivals) and F_{dep} (departures)
- $S \subseteq F_{arr} \times F_{dep}$: Set of spin turnaround flights (a, d) where a is the arriving flight and d is the departing flight using the exact same physical aircraft on the same stand.
- R : Set of all physical vehicles and candidate vehicles available to the model.
 - $R^{amb} = \{r \in R: type_r = Amb\}$.
 - $R^{mini} = \{r \in R: type_r = Mini\}$.
 - $R^{future} = \{r \in R: isFuture_r = 1\}$.
- M : Set of mutually exclusive operational service modes, m :
 - $m = SA$: Solo Ambulift (Ambulift handles both vertical drop and horizontal transit).
 - $m = CM$: Combined Minibus (Ambulift handles vertical drop; Minibus handles horizontal drive)
 - $m = CP$: Combined Pusher (Ambulift handles vertical drop; Pusher agent walks with passenger)
 - $m = SM$: Solo Minibus (No vertical drop required; Minibus handles horizontal drive).
 - $m = SP$: Solo Pusher (No vertical drop required; Pusher agent walks with passenger).

2. Pre-processed Input data

The following fields are created upstream before the optimisation model is built

Flight Identity and Timing Inputs

For each flight $f \in F$, the model requires:

- $flightKey_f$: unique flight identifier
- $scheduledTime_f$: scheduled flight time
- $serviceAnchorTime_f$: fixed service anchor time to construct service intervals
 - For arrivals the service anchor time is treated as the beginning of the relevant service process
 - For departures, the service anchor is treated as the point by which the relevant service process must complete.
- $A_f = minutes(serviceAnchorTime_f - origin)$ in minutes
- $Arrival_f = \begin{cases} 1, & \text{if flight } f \text{ is an arrival} \\ 0, & \text{if flight } f \text{ is a departure} \end{cases}$

Demand inputs

- P_f : total PRM count on the flight
- D_f^{WCHC} : number of WCHC passengers
- D_f^{WCHS} : number of WCHS passengers
- D_f^{WCHR} : number of WCHR passengers
- D_f^{seat} : number of PRM passengers requiring standard vehicle seats

- D_f^{wc} : number of PRM passengers requiring wheelchair tie-down spaces
- D_f^{vert} : the total number of passengers requiring a vertical transfer
- $D_f^{vert,seat}$: number of vertical-transfer passengers requiring standard vehicle seats
- $D_f^{vert,wc}$: number of vertical-transfer passengers requiring wheelchair tie-down spaces

Within the model, the demand values will be used differently depending on the service mode:

- In *SA*, the Ambulift is treated as handling the vertical and horizontal movement, so Ambulift capacity is checked against total seat and wheelchair demand
- In *CM* and *CP*, the Ambulift is used only for the vertical component, so Ambulift capacity is checked against vertical seat and vertical wheelchair demand
- In *CM* and *SM*, minibus capacity is checked against total seat and wheelchair demand.

Operational Flags

For each flight f :

- $Remote_f \in \{0,1\}$: indicates that the flight is on a remote stand
- $EffectiveRemote_f \in \{0,1\}$: indicates that the flight is effectively remote or requiring a vertical transfer
- $NeedVertical_f \in \{0,1\}$: indicates the flight must use a vertical-service mode
- $Domestic_f \in \{0,1\}$: Flag indicating the flight belongs to the domestic sector
- $Safety_f \in \{0,1\}$: Flag indicating the flight is on a designated safety stand

Spin Turnaround Processing

For each flight f :

- $SpinArrival_f \in \{0,1\}$: indicates that an arrival flight is part of a spin turnaround process
- $SpinCandidate_f \in \{0,1\}$: indicates that a departure flight is a spin departure candidate

3. Model Parameters

Vehicle Parameters

- $type_r \in \{Amb, Mini\}$
- cap_r^{seat} : standard passenger seating capacity
- cap_r^{wc} : wheelchair tie-down capacity
- $capex_r$: cost or preference weight associated with the vehicle
- $isFuture_r = 1$: indicates a future purchase candidate
- $baseModel_r$ identifies the underlying model type

Future Ambulifts are not permitted, therefore:

$$R^{future} \cap R^{amb} = \emptyset$$

Duration Parameters (fixed assumptions)

- τ_{amb}^{solo} : duration for a Solo Ambulift run
- τ_{amb}^{comb} : duration for the vertical component of a combined job
- τ_{mini} : duration for a minibus transit
- τ_{push} : duration for a walking pusher agent assignment
- β : Handover buffer time between the Ambulift vertical process and the onward horizontal movement by minibus or pusher

Spin- specific duration parameters

- $\tau_{amb}^{spin,arr}$: duration for a spin arrival Ambulift process
- $\tau_{amb}^{spin,dep}$: duration for a spin departure-candidate ambulift process

SLA Parameters

- T_{SLA} : Target SLA window duration (e.g., 20 minutes)

4. Trip Counts and Service Durations

For each flight $f \in F$, ambulift $r \in R^{amb}$, and vertical service mode $m \in M^{vert}$, the vehicle passenger demand that must be carried by the Ambulift is:

- $Q_{f,SA}^{amb,seat} = D_f^{seat}$
- $Q_{f,SA}^{amb,wc} = D_f^{wc}$

This is because Ambulift handles both vertical and horizontal movements

- $Q_{f,SA}^{amb,seat} = D_f^{vert,seat} \quad m \in \{CM, CP\}$
- $Q_{f,SA}^{amb,wc} = D_f^{vert,wc} \quad m \in \{CM, CP\}$

This is because the ambulift only handles the vertical component.

The number of Ambulift trips required is:

$$Trips_{f,r,m}^{amb} = \max \left(1, \left\lceil \frac{Q_{f,m}^{amb,seat}}{cap_r^{seat}} \right\rceil, \left\lceil \frac{Q_{f,m}^{amb,wc}}{cap_r^{wc}} \right\rceil \right)$$

The Ambulift trip duration is:

$$\begin{cases} \tau_{amb}^{spin,arr} & Arrival_f = 1 \wedge SpinArrival_f = 1 \\ \tau_{amb}^{spin,dep} & Arrival_f = 0 \wedge SpinCandidate_f = 1 \\ \tau_{amb}^{solo}, & m = SA \\ \tau_{amb}^{comb}, & m \in \{CM, CP\} \end{cases}$$

The total Ambulift assignment duration is:

$$Dur_{f,r,m}^{amb} = Trips_{f,r,m}^{amb} \cdot \tau_{f,m}^{amb}$$

The number of concurrent minibuses required to satisfy total flight passenger demand:

$$K_{f,Mini} = \max \left(\left\lceil \frac{D_{f,seat}}{Cap_{Minibus,seat}} \right\rceil, \left\lceil \frac{D_{f,wc}}{Cap_{Minibus,wc}} \right\rceil \right)$$

5. Precomputed Service Intervals

Each interval uses half-open time convention $[start, end)$. Two intervals overlap if $s_1 < e_2$ and $s_2 < e_1$

Ambulift Intervals

For each flight $f \in F$, ambulift $r \in R^{amb}$, and vertical service mode $m \in M^{vert}$, the Ambulift interval is:

$$\begin{aligned} I_{f,r,m}^{amb} &= [s_{f,r,m}^{amb}, e_{f,r,m}^{amb}) \\ s_{f,r,m}^{amb,arr} &= A_f \\ e_{f,r,m}^{amb,arr} &= A_f + Dur_{f,r,m}^{amb} \\ e_{f,r,m}^{amb,dep} &= A_f \\ s_{f,r,m}^{amb,dep} &= A_f - Dur_{f,r,m}^{amb} \end{aligned}$$

Where $serviceAnchorTime_f$ is in minutes

Minibus Intervals

For each flight $f \in F$, minibus $r \in R^{mini}$, and mini-bus supported service mode $m \in M^{mini}$, the minibus interval is:

$$\begin{aligned} I_{f,r,m}^{mini} &= [s_{f,r,m}^{mini}, e_{f,r,m}^{mini}) \\ s_{f,r,SM}^{mini,arr} &= A_f \\ e_{f,r,SM}^{mini,arr} &= A_f + \tau_{mini} \\ e_{f,r,SM}^{mini,dep} &= A_f \\ s_{f,r,SM}^{mini,dep} &= A_f - \tau_{mini} \end{aligned}$$

For combined minibus mode, the minibus interval is positioned relative to the latest possible Ambulift completion time for the flight. This ensures the minibus remains available throughout the handover period and subsequent transit.

First, we define:

$$MaxDur_f^{comb} = \max_{r \in R^{amb}} Dur_{f,r,CM}^{amb}$$

For arrivals:

$$\begin{aligned} s_{f,CM}^{mini} &= A_f + MaxDur_f^{comb} - \beta \\ e_{f,CM}^{mini} &= A_f + MaxDur_f^{comb} - \tau_{mini} \end{aligned}$$

For departures:

$$e_{f,CM}^{mini} = A_f$$

$$s_{f,CM}^{mini} = A_f - (\beta + \tau_{mini})$$

Staff Intervals

For each flight $f \in F$ and mode $m \in M$, the model derives a staff interval:

$$I_{f,m}^{staff} = [s_{f,m}^{staff}, e_{f,m}^{staff})$$

The staff duration by mode is:

$$Dur_{f,SA}^{staff} = \max_{r \in R^{amb}} Dur_{f,r,SA}^{amb}$$

$$Dur_{f,CM}^{staff} = MaxDur_f^{comb} + \beta + \tau_{mini}$$

$$Dur_{f,CP}^{staff} = MaxDur_f^{comb} + \beta + \tau_{push}$$

$$Dur_{f,SM}^{staff} = \tau_{mini}$$

$$Dur_{f,SP}^{staff} = \tau_{push}$$

For arrivals:

$$s_{f,m}^{staff} = A_f$$

$$e_{f,m}^{staff} = A_f + Dur_{f,m}^{staff}$$

For departures:

$$e_{f,m}^{staff} = A_f$$

$$s_{f,m}^{staff} = A_f - Dur_{f,m}^{staff}$$

6. Arrival SLA Logic

EDI SLA is that arrival PRMs should receive first contact with 20 minutes, 98% of the time.

Vertical Mode First-Contact

For each arrival flight $f \in F^{arr}$, Ambulift $r \in R^{amb}$, and vertical mode $m \in M^{vert}$, the start time of the Ambulift trip k is:

$$StartTrip_{f,r,m,k} = (k - 1)\tau_{f,m}^{amb} \quad \text{where } k = 1, \dots, Trips_{f,r,m}^{amb}$$

A trip is on time if:

$$StartTrip_{f,r,m,k} \leq T^{SLA}$$

The number of on-time Ambulift trips is:

$$OnTimeTrips_{f,r,m} = |\{k: StartTrip_{f,r,m,k} \leq T^{SLA}\}|$$

The on-time seated and wheelchair capacity are:

$$OnTimeSeatCap_{f,r,m} = OnTimeTrips_{f,r,m} \cdot cap_r^{seat}$$

$$OnTimeWcCap_{f,r,m} = OnTimeTrips_{f,r,m} \cdot cap_r^{wc}$$

The number of PRMs contacted on time is:

$$OnTimeSeat_{f,r,m} = \min(Q_{f,m}^{amb,seat}, OnTimeSeatCap_{f,r,m})$$

$$OnTimeWc_{f,r,m} = \min(Q_{f,m}^{amb,wc}, OnTimeWcCap_{f,r,m})$$

The estimated Late PRMs for an Ambulift-supported mode are:

$$LatePRM_{f,r,m}^{amb} = (0, Q_{f,m}^{amb,seat} + Q_{f,m}^{amb,wc} - OnTimeSeat_{f,r,m} - OnTimeWc_{f,r,m})$$

For vertical modes, the selected late PRM value is:

$$LatePRM_{f,m} = \sum_{r \in R^{amb}} LatePRM_{f,r,m}^{amb} \cdot amb_{f,r,m} \quad \forall f \in F^{arr}, \quad m \in \{SA, CM, CP\}$$

For non-vertical modes, first contact is represented by the start of the minibus or pusher service interval.

$$ContactTime_{f,SM} = s_{f,SM}^{mini} - serviceAnchorTime_f$$

$$ContactTime_{f,SP} = s_{f,SP}^{staff} - serviceAnchorTime_f$$

Then:

$$LatePRM_{f,SM} = \begin{cases} 0, & ContactTime_{f,SM} \leq T^{SLA} \\ P_f, & ContactTime_{f,SM} > T^{SLA} \end{cases}$$

$$LatePRM_{f,SP} = \begin{cases} 0, & ContactTime_{f,SP} \leq T^{SLA} \\ P_f, & ContactTime_{f,SP} > T^{SLA} \end{cases}$$

The total late arrival PRMs are:

$$LateArrivalPRMs = \sum_{f \in F^{arr}} \sum_{m \in M} LatePRM_{f,m} \cdot x_{f,m}$$

The total arrival PRMs are:

$$ArrivalPRMs = \sum_{f \in F^{arr}} P_f$$

The achieved arrival SLA is:

$$ArrivalSLA\% = 100 \left(1 - \frac{LateArrivalPRMs}{ArrivalPRMs} \right)$$

The SLA target condition is $ArrivalSLA\% \geq 98$ and therefore the constraint should be $LateArrivalPRMs \leq (1 - TargetPct) \cdot ArrivalPRMs$

7. Optimisation Decision Variables

- $x_{f,m} \in \{0,1\}$: Equals 1 if flight f is serviced utilising operational mode $m \in M$; 0 otherwise.
- $amb_{f,r,m} \in \{0,1\}$: Equals 1 if ambulift r is assigned to flight f in mode m , 0 otherwise. Where $m \in M^{vert}$
- $mini_{f,r,m} \in \{0,1\}$: Equals 1 if miniubus r is assigned to flight f in mode m , 0 otherwise. Where $m \in M^{mini}$
- $use_r \in \{0,1\}$: Equals 1 if vehicle r is used at least once, 0 otherwise.
- $buy_r \in \{0,1\}$: Equals 1 if future candidate vehicle r is purchased, 0 otherwise. Where $r \in R^{future}$

Shortfall variables

To avoid infeasibility when the current and future candidate fleet cannot fully cover all requirements.

For each flight $f \in F$ and vertical mode $m \in M^{vert}$:

$$short_{f,m}^{amb} \geq 0$$

For each flight $f \in F$ and minibus-supported mode $m \in M^{mini}$:

$$short_{f,m}^{seat} \geq 0$$

$$short_{f,m}^{wc} \geq 0$$

8. Derived Reporting Quantities

Maintained Fleet Quantity

$$N_v = CurrentAvailable_v + FutureBought_v$$

Where:

$$CurrentAvailable_v = \sum_{r \in R_v} (1 - isFuture_r)$$

$$FutureBought_v = \sum_{r \in R_v \cap R^{future}} buy_r$$

$$R_v = \{r \in R: group_r = v\}$$

Peak Staff

$$N_{staff}^{peak} = \max_{\tau \in E^{staff}} \sum_{f \in F} StaffReq_{f,m_f} \cdot (\tau \in I_{f,m_f}^{staff})$$

Where:

$$m_f = \{m \in M : x_{f,m} = 1\}$$

E^{staff} is set of start and end event points from chosen staff intervals

9. Service Mode Constraints

One Mode per Flight: Every Flight must be serviced by exactly one operational framework:

$$\sum_{m \in M} x_{f,m} = 1 \quad \forall f \in F$$

Vertical Requirement Alignment: The chosen operational routing (m) must strictly align with the passenger-driver vertical requirement metric:

$$\sum_{m \in \{SA, CM, CP\}} x_{f,m} = NeedVertical_f \quad \forall f \in F$$

$$\sum_{m \in \{SM, SP\}} x_{f,m} = 1 - NeedVertical_f \quad \forall f \in F$$

Pusher Mode Stand Restriction: Pusher Agents cannot operate on remote stands where walking to the terminal is impossible:

$$x_{f,CP} + x_{f,SP} \leq 1 - Remote_f \quad \forall f \in F$$

Domestic Arrival Ambulift Exclusion: Ambulifts are logistically banned from performing horizontal and arrival movements for domestic sectors due to customs/terminal design constraints:

$$x_{f,SA} \leq 1 - (Arrival_f \cdot Domestic_f) \quad \forall f \in F$$

Safety Stand Pusher Restrictions: Combined Pusher agents are prohibited from operating on safety stands. Solo pushers are permitted.

$$\sum_{m \in \{CP\}} x_{f,m} \leq 1 - Safety_f \quad \forall f \in F$$

10. Vehicle Assignment and Capacity Constraints

Ambulift Assignment for Vertical Modes: If a vertical mode is selected, one Ambulift must be assigned unless the model uses an Ambulift shortfall:

$$\sum_{r \in R^{amb}} amb_{f,r,m} + short_{f,m}^{amb} = x_{f,m} \quad \forall f \in F, m \in M^{vert}$$

Minibus Seat Capacity: For minibus-supported modes, assigned minibus seated capacity must cover seated demand unless a seated-capacity shortfall is used:

$$\sum_{r \in R^{mini}} cap_r^{seat} \cdot mini_{f,r,m} + short_{f,m}^{seat} \geq D_{f,m}^{seat} \cdot x_{f,m} \quad \forall f \in F, m \in M^{mini}$$

Minibus Wheelchair Capacity: For minibus-supported modes, assigned minibus wheelchair capacity must cover wheelchair demand unless a wheelchair-capacity shortfall is used:

$$\sum_{r \in R^{mini}} cap_r^{wc} \cdot mini_{f,r,m} + short_{f,m}^{wc} \geq D_{f,m}^{wc} \cdot x_{f,m} \quad \forall f \in F, m \in M^{mini}$$

11. Vehicle Use and Future Purchase Constraints

Ambulift Use Activation: If an Ambulift is assigned to a flight, it must be marked as used:

$$amb_{f,r,m} \leq use_r \quad \forall f \in F, r \in R^{amb}, m \in M^{vert}$$

Minibus Use Activation: If a minibus is assigned to a flight, it must be marked as used:

$$mini_{f,r,m} \leq use_r \quad \forall f \in F, r \in R^{mini}, m \in M^{mini}$$

Future Minibus Purchase Requirement: A future minibus candidate can only be assigned if purchased:

$$mini_{f,r,m} \leq buy_r \quad \forall f \in F, r \in R^{future}, m \in M^{mini}$$

Future Vehicle Use Requires Purchase: A future minibus candidate can only be marked as used if it is purchased

$$use_r \leq buy_r \quad \forall r \in R^{future}$$

12. Timeline No-overlap Constraints

Prevention of the same physical vehicle being assigned to overlapping service intervals.

Ambulift Overlap Constraints

Let Ω^{amb} be the set of all overlapping Ambulift assignment pairs:

$$(f_1, r, m_1, f_2, r, m_2) \in \Omega^{amb}$$

Where:

$$I_{f_1, r, m_1}^{amb} \cap I_{f_2, r, m_2}^{amb} \neq \emptyset$$

$$f_1 \neq f_2$$

For every overlapping pair:

$$amb_{f_1, r, m_1} + amb_{f_2, r, m_2} \leq 1 \quad \forall (f_1, r, m_1, f_2, r, m_2) \in \Omega^{amb}$$

Minibus Overlap Constraints

Let Ω^{mini} be the set of all overlapping minibus assignment pairs:

$$(f_1, r, m_1, f_2, r, m_2) \in \Omega^{mini}$$

Where:

$$I_{f_1, r, m_1}^{mini} \cap I_{f_2, r, m_2}^{mini} \neq \emptyset$$

$$f_1 \neq f_2$$

For every overlapping pair:

$$mini_{f_1, r, m_1} + mini_{f_2, r, m_2} \leq 1 \quad \forall (f_1, r, m_1, f_2, r, m_2) \in \Omega^{mini}$$

13. Staffing Logic

For each flight f and mode m , define:

$$StaffReq_{f,m}$$

The staff requirement by mode is:

$$StaffReq_{f,SA} = Staff_{amb}$$

$$StaffReq_{f,SM} = Staff_{mini}$$

$$StaffReq_{f,CM} = Staff_{amb} + Staff_{mini}$$

$$StaffReq_{f,CP} = Staff_{amb} + StaffPerPRM_{push} \cdot P_f$$

$$StaffReq_{f,SP} = StaffPerPRM_{push} \cdot P_f$$

The staff-minutes associated with each flight-mode option are:

$$StaffMinutes_{f,m} = StaffReq_{f,m} \cdot Dur_{f,m}^{staff}$$

14. Objective Function

Minimise:

$$\min Z = Z^{short} + Z^{buy} + Z^{use} + Z^{staff} + Z^{assign} + Z^{late}$$

Where:

$$Z^{short} = \lambda_{short} \left[\sum_{f \in F} \sum_{m \in M^{vert}} short_{f,m}^{amb} + \sum_{f \in F} \sum_{m \in M^{mini}} short_{f,m}^{seat} + \sum_{f \in F} \sum_{m \in M^{mini}} short_{f,m}^{wc} \right]$$

$$Z^{buy} = \lambda_{buy} \sum_{r \in R^{future}} capex_r \cdot buy_r$$

$$Z^{use} = \lambda_{use} \sum_{r \in R} capex_r \cdot use_r$$

$$Z^{staff} = \lambda_{staff} \sum_{f \in F} \sum_{m \in M} StaffMinutes_{f,m} \cdot x_{f,m}$$

$$Z^{assign} = \lambda_{assign} \left[\sum_{f \in F} \sum_{r \in R^{amb}} \sum_{m \in M^{vert}} amb_{f,r,m} + \sum_{f \in F} \sum_{r \in R^{mini}} \sum_{m \in M^{mini}} mini_{f,r,m} \right]$$

$$Z^{late} = \lambda_{late} \sum_{f \in F^{arr}} \sum_{m \in M} LatePRM_{f,m} \cdot x_{f,m}$$